Akhila Pasupunuri

425-773-1990 | akhilapnuri@gmail.com | linkedin.com/in/akhila-pasupunuri | github.com/akhilapnuri

Education

University of Maryland College Park, MD

B.S. in Computer Science (Machine Learning Specialization), Minor in Mathematics *Aug. 2025 – May 2028*

- **Relevant Coursework:** Object-Oriented Programming, Computer & Operating Systems, Discrete Mathematics, Programming Languages (OCaml, Rust), Algorithms, Data Science, Linear Algebra, Calculus I–III

Experience

Undergraduate Researcher Nov. 2026 – Present

University of Maryland – NIH-Funded AI Chatbot Safety Project *College Park, MD*

- Designing a **Graph RAG** retrieval system over a validated corpus of CDC, NIH, and NCI documents for an LLM chatbot that supports parents' HPV vaccine decisions, advised by Dr. Philip Resnik and Dr. Xiaoli Nan

Software Engineer Oct. 2026 – Present

Hack4Impact – Client: Rockville Makerspace *College Park, MD*

- Building a member check-in app for Rockville Makerspace with **React**, **TypeScript**, **tRPC**, and **Firebase**

Founding Engineer Intern Jun. 2026 – Aug. 2026

SheRecs (Early-Stage Startup) *Seattle, WA*

- Built a personalized recommender using **pgvector**, **MiniLM** embeddings, and cosine-similarity KNN search to raise engagement with suggested recommendations
- Co-built AI search using **Claude Haiku** query expansion with **Redis** caching and daily quotas, reducing zero-result searches while keeping API costs low
- Co-built a friends-first social network with **Next.js** and **PostgreSQL**, adding rate limiting across 65 endpoints and launching to 402 beta users

Software Engineer Feb. 2026 – May 2026

Hack4Impact – Client: Kisses for Kyle Foundation *College Park, MD*

- Built a donor checkout flow with **React**, **TypeScript**, and **Firebase Auth**, so only verified donors can claim gifts for 600 families
- Fixed an N+1 query problem with batched **Firestore** queries, cutting database round trips by about 90%
- Secured server functions with token validation and access checks, and built staff dashboards to track gift status across all families

iOS Engineer Sep. 2025 – Feb. 2026

App Dev Club – Client: University of Maryland *College Park, MD*

- Built a Monte Carlo net-worth simulator in **Flask**, **NumPy**, and **pandas** for UMD's Financial Wellness Center, projecting 10–60 year outcomes across 470+ careers and 51 states
- Designed 15 REST endpoints across 7 Flask blueprints with **SQLAlchemy** and **Alembic**, powering daily finance quizzes, leaderboards, course recommendations, and prerequisite planning
- Automated a daily **BeautifulSoup** scraper with stable UUID deduplication to sync Smith School events into the app, replacing manual event entry

Projects

Recall | *Bitcamp Winner – Bloomberg's Most Philanthropic Hack* Apr. 2026

- Built a **SwiftUI** iOS app and a **Flask** REST API (7 endpoints) that identify faces in real time using **InsightFace** ArcFace 512-d embeddings with a 0.4 cosine-similarity threshold, helping dementia patients recognize family members through spoken name announcements
- Recorded and summarized patient conversations on-device using iOS Speech Recognition and a local **Mistral 7B** model (**Ollama**), keeping all audio and transcripts off third-party servers

Technical Skills

Languages: Python, Java, C, C++, TypeScript, JavaScript, Swift, SQL, OCaml, Rust
Frameworks: React, Next.js, Flask, SwiftUI, ARKit, tRPC
Databases: SQL (PostgreSQL, pgvector, Supabase), NoSQL (Firestore, Redis)
Cloud & DevOps: AWS, GCP, Microsoft Azure, Firebase, CI/CD (GitHub Actions), Git
AI/ML & Libraries: NumPy, pandas, SQLAlchemy, InsightFace, MiniLM, Ollama, Claude API, BeautifulSoup